CIE AS PHYSICS (9702) • QUESTION PAPER

10.1 Practical Circuits

Syllabus Section: 10. D.C. Circuits **Paper Type:** Structured Questions (Paper 2) **Total:** 5 Questions (60 Marks)

Question 1 Easy [9 marks]

Part (a) [2 marks]

Explain why the electromotive force (e.m.f) of a cell with internal resistance may be more than the terminal potential difference (p.d.) of the cell.

Part (b) [3 marks]

Compare the definitions of e. m .f and p. d.

Part (c) [1 mark]

For a cell, explain the term *internal resistance*.

Part (d) [3 marks]

A battery of e.m.f 15.2 V and internal resistance *r* is connected to a resistor A, as shown in Fig 1.1.

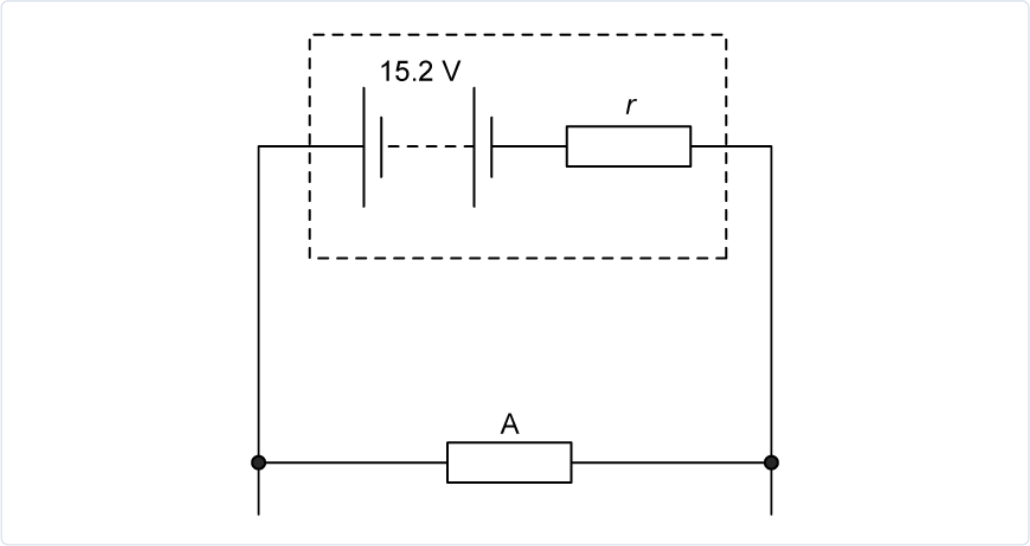


Fig 1.1

The current in resistor A is 400 mA and the power dissipated by it is 6 W.

Calculate *r* of the battery.

Question 2 Easy [14 marks]

Part (a) [3 marks]

The electromotive force, E is defined by the equation

$$E = I\,(R + r)$$

State the definition of the following variables and an appropriate unit for each.

(i) I

[1]

(ii) R

[1]

(iii) r

[1]

Part (b) [5 marks]

A circuit connecting a cell with internal resistance to a resistor is shown in Fig. 1.1.

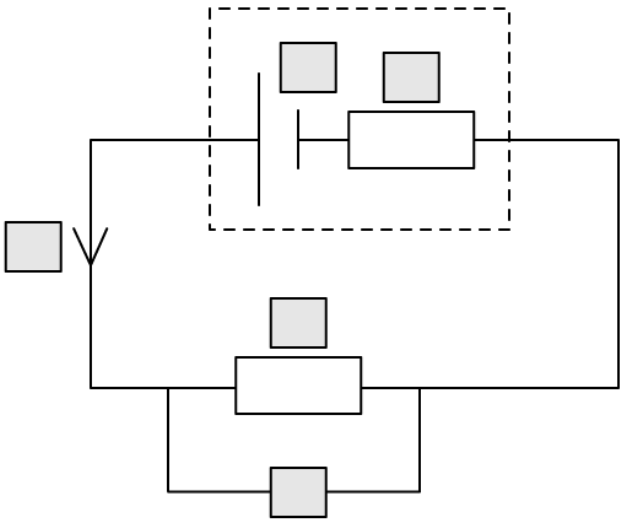


Fig. 1.1

Place in each of the grey boxes one of the following variables:

V	***r***	***E***	***I***	***R***

Part (c) **[3 marks]**

A current of 0.6 mA flows through the circuit in Fig. 1.1 and the resistor has a resistance of 2.5 kΩ.

Calculate the terminal potential difference of the circuit.

Part (d) **[3 marks]**

The internal resistance of the cell is 1.3 kΩ.

Calculate the emf of the cell using the equation E.m.f. $E = I(R + r)$.

Question 3 Medium [13 marks]

Part (a) [3 marks]

Explain why the terminal potential difference (p.d) of a cell with internal resistance may differ from the electromotive force (e.m.f) of the cell.

Part (b) [5 marks]

A battery of e.m.f 6.2 V and internal resistance 7.0 Ω is connected in series with a resistor of resistance 11 Ω as shown in Fig 1.1.

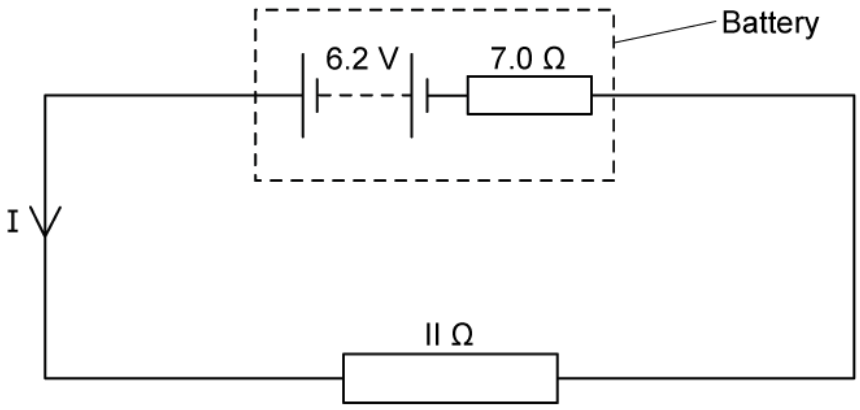


Fig. 1.1

The current I flows through the circuit

Determine

(i) the current I in the circuit,

[3]

(ii) the terminal p.d. of the battery.

[2]

Part (c) [2 marks]

A second identical resistor is added in series with the resistor in Fig 1.1.

Quantitatively explain the change in the current in the circuit.

Part (d) [3 marks]

The second 11 Ω resistor is now connected in parallel to the initial 11 Ω resistor in Fig 1.1.

Qualitatively explain the effect this has on the battery.

Question 4 Medium [13 marks]

Part (a) [3 marks]

Two circuits X and Y shown in Fig 1.1 were used by a student to test a battery of four identical cells. In circuit X the battery is connected to a voltmeter registering 12 V and there was no load resistor. In circuit Y a load resistor of 5.2 Ω is connected in parallel with an ammeter registering 0.41 A and a voltmeter registering 4.9 V. Assume that the meters in the circuits are ideal.

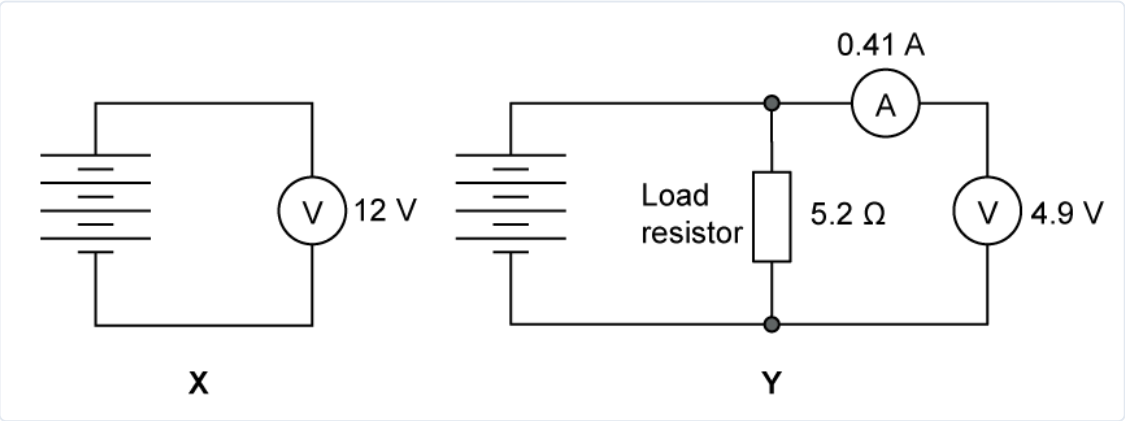


Fig. 1.1

Explain why there is a difference in voltages recorded in the two circuits.

Part (b) [3 marks]

Calculate the internal resistance of a single cell from Fig. 1.1.

Part (c) [3 marks]

One of the cells in the battery is reversed.

Determine the new reading on the ammeter in circuit Y.

Part (d) [4 marks]

In circuit Y, the resistance of the load resistor R is altered so that a series of values on the voltmeter and the corresponding values of the current on the ammeter are obtained.

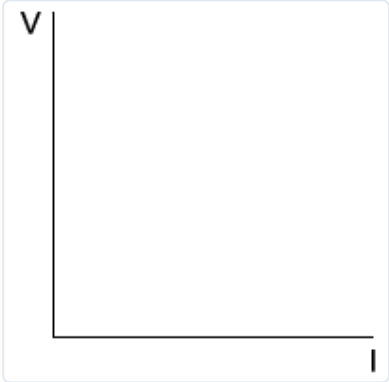


Fig. 1.2

Using the axes in Fig.1.2, sketch the graph you would expect to obtain as R is changed. State how the values of ε and r can be obtained from the graph.

Question 5 Medium [11 marks]

Part (a) [4 marks]

A battery of e.m.f. 60.0 V and internal resistance, *r*, is connected in a circuit as shown in Fig. 1.1.

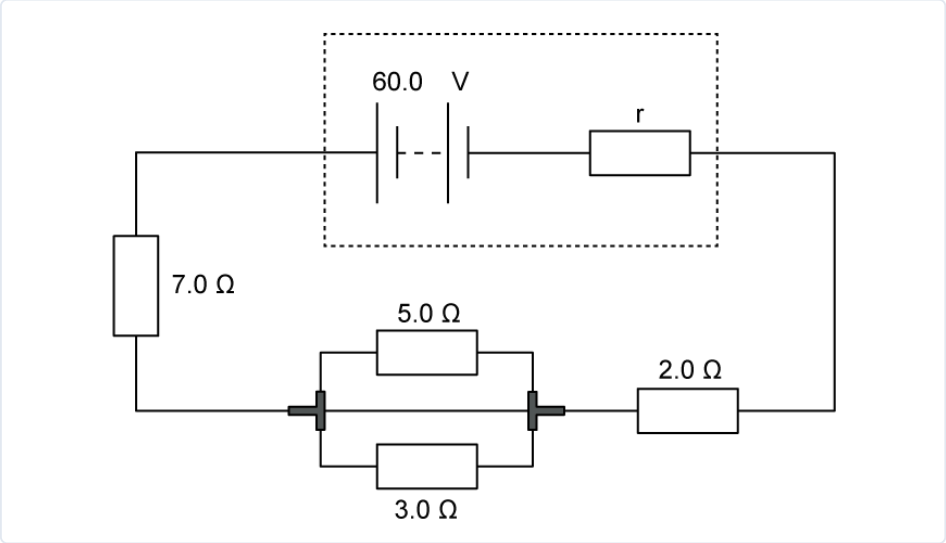


Fig 1.1

The current in the battery is 2.3 A.

Calculate the internal resistance *r*.

Part (b) [2 marks]

Calculate the energy dissipated in the battery in 4.0 minutes.

Part (c) [2 marks]

Calculate the percentage of the total energy transformed by the battery that is dissipated in the battery in 4.0 minutes.

Part (d) [3 marks]

The internal resistance of the battery affects the efficiency of the transfer of energy from the battery to the circuit.

Explain what causes internal resistance and why this affects the efficiency.